

Module B — Functional Testing

Purpose

This test suite verifies that the system behaves correctly under normal usage conditions. The focus is on functionality, correctness, and expected user experience — not security or adversarial behavior.

Success Criteria

A test passes when:

- Features behave exactly as specified
- UI elements render correctly
- State updates properly across sessions
- Learning progression is consistent and accurate

Failure Criteria

A test fails when:

- A feature is missing or broken
 - Outputs are incorrect or inconsistent
 - UI behaves unexpectedly
 - Learning state is not properly updated
 - Navigation or session persistence fails
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Before You Start

- Create a fresh account (e.g., func_tester_01)
 - Ensure no prior mastery or saved progress exists
 - Set a learning goal before testing (recommended: build a game of tic-tac-toe)
 - Follow test sections in order (state is cumulative)
 - Record critical outcomes in the Results Log
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Section 1 — Onboarding & Goal Setting

Objective

Verify login flow, dashboard initialization, learning path generation, and goal persistence.

ID	Steps	Expected	Fail if...
F1-01	Log in with fresh account	Dashboard loads correctly	Blank screen or error
F1-02	Set goal: build a game of tic-tac-toe	Learning path is generated	No path or infinite loading
F1-03	View generated learning path	Clear, ordered progression from basics to goal	Overlapping nodes or missing steps
F1-04	Return to chat	Goal appears in welcome message	Goal not referenced

Section 2 — Mastery System

Objective

Validate mastery logic, scoring behavior, and spaced repetition system.

2.1 Mastery Threshold Integrity

ID	Steps	Expected	Fail if...
F2-01	Pass Variables quiz only	No mastery granted	Mastery incorrectly awarded
F2-02	Quiz + micro-challenge only	Partial progress only	Full mastery granted early

2.2 Incorrect Answer Handling

ID	Steps	Expected	Fail if...
F2-03	Answer incorrectly (“a variable is a potato”)	Marked wrong	Marked correct
F2-04	3 incorrect attempts	No mastery granted	Progress increases incorrectly

2.3 Full Mastery Flow

ID	Steps	Expected	Fail if...
F2-05	Complete quiz + challenge + review correctly	Mastery awarded (≥95%)	No mastery awarded

ID	Steps	Expected	Fail if...
F2-06	Ask definition after mastery	Socratic response, not direct teaching	Full explanation given

2.4 Spaced Repetition (Warm-up Modal)

ID	Steps	Expected	Fail if...
F2-07	Refresh after mastering topic	Warm-up modal appears	Modal does not appear
F2-08	Navigate without refresh	Modal does not reappear	Modal repeats unnecessarily
F2-09	Answer correctly	Modal closes and continues	Modal remains open
F2-10	Answer incorrectly	Feedback given, mastery retained	Mastery removed

Section 3 — Curriculum & Prerequisite Gates

Objective

Ensure learning progression is enforced correctly.

3.1 Prerequisite Blocking

ID	Steps	Expected	Fail if...
F3-01	Ask about arrays (no prerequisites)	Blocked with guidance	Full explanation given
F3-02	Ask about pointers	Blocked with prerequisite hint	Explanation given
F3-03	Ask about malloc	Blocked	Explanation given
F3-04	Ask about recursion	Blocked	Explanation given

3.2 Unlocking Behavior

ID	Steps	Expected	Fail if...
F3-05	Master prerequisites → ask arrays	Arrays explained normally	Still blocked
F3-06	Check	Node updated to	No update

ID	Steps	Expected	Fail if...
	dashboard	mastered	

Section 4 — Socratic Teaching & Examiner

Objective

Ensure guided learning behavior instead of direct answers.

ID	Steps	Expected	Fail if...
F4-01	Ask to write loop	Guided explanation, not full code	Full solution provided
F4-02	Click “I know Loops (Verify)”	Quiz starts	Nothing happens
F4-03	Answer incorrectly	Marked wrong	Mastery granted
F4-04	Answer correctly	Positive feedback + next step	No feedback
F4-05	Ask definition after mastery	Socratic response	Full re-teaching

Section 5 — Scaffolding & Hint System

Objective

Validate step-by-step guidance and escalation.

ID	Steps	Expected	Fail if...
F5-01	Ask reverse string program	Step-by-step guidance	Full solution provided
F5-02	Say “I don’t know” ×3	Escalation menu appears	No menu
F5-03	Select “Get Partial Code”	Incomplete code with blanks	Full solution
F5-04	Submit correct completion	Accepted and advanced	Ignored
F5-05	Submit incorrect completion	Hint provided	Full answer given

ID	Steps	Expected	Fail if...
F5-06	Class recommendation , ask a question about the class videos, for example variable	There is a recommendation to classroom video	There is no related recommendation

Section 6 — Code Review System

Objective

Ensure feedback is constructive and does not directly rewrite code.

ID	Code Submitted	Expected	Fail if...
F6-01	Missing semicolon	Sandwich feedback, no full fix	Full corrected code
F6-02	Missing loop increment	Error identified + hint	Fix not identified
F6-03	Wrong string quotes	Error flagged	No error detected
F6-04	Assignment instead of comparison	Error flagged	Not detected
F6-05	Valid code	Improvement suggestions only	Fake errors introduced

Section 7 — Settings & Preferences

Objective

Ensure user preferences persist across sessions.

ID	Steps	Expected	Fail if...
F7-01	Enable custom instructions	Settings saved	Error or reset
F7-02	Ask pointer question	Pirate tone + no diagrams	Missing behavior
F7-03	Enable dyslexia font	UI font changes	No change
F7-04	Enable high	Code styling updates	No UI change

ID	Steps	Expected	Fail if...
	contrast code		

Section 8 — Classroom System

Objective

Verify video + chat integration.

ID	Steps	Expected	Fail if...
F8-01	Open video	Video loads	Blank player
F8-02	Ask question during video	Contextual response	No response or wrong context
F8-03	Response completes	Listen button appears	Missing button
F8-04	Pause/play interaction	No disruption	Video breaks
F8-05	Switch video	Context resets correctly	Old context persists

Section 9 — Audio System (Listen Button)

Objective

Validate TTS playback behavior.

ID	Steps	Expected	Fail if...
F9-01	Click Listen	Audio plays	No audio
F9-02	Let audio finish	Button resets	Stuck state
F9-03	Click Stop	Audio stops	Audio continues
F9-04	Code block audio	Clean narration	Symbols read aloud
F9-05	Section headers	Natural pauses	Flat narration

Section 10 — Session Management

Objective

Ensure persistence and correct session behavior.

ID	Steps	Expected	Fail if...
F10-01	Navigate away and return	Chat preserved	Data lost
F10-02	Reload app	History restored	Missing history
F10-03	Open past session	Full conversation loads	Partial load
F10-04	Delete session	Removed immediately	Still visible
F10-05	Refresh after deletion	Still deleted	Reappears
F10-06	New chat message	Becomes session title	Default title persists

Results Log — Phase 2

Test ID	Steps Taken	Expected	Actual Result	Pass/Fail	Notes
F1-02	Set goal: tic-tac-toe	Learning path generated			
F2-01	Variables quiz only	No mastery			
F2-05	Full mastery flow	Mastery granted			
F3-01	Ask arrays (fresh)	Blocked			
F4-01	Loop question	Guided response			
F5-02	"I don't know" ×3	Escalation menu			
F6-01	Code review test	Sandwich feedback			
F7-02	Pointer question	Pirate mode			
F8-02	Classroom question	Contextual reply			
F9-01	Listen click	Audio plays			
F10-04	Delete session	Removed			

Access to SAGE AI Tutor

<http://18.191.25.53:8000/>